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SecureVote: Blockchain Voting System for a Transparent Future

Abstract

In a world where trust in digital systems is paramount, SecureVote redefines democracy on campus. This blockchain-based voting platform ensures every vote is secure, transparent, and verifiable, empowering PSP students to shape their future with confidence.

Introduction

Traditional voting systems are vulnerable to tampering and lack transparency. SecureVote uses blockchain technology to create a decentralized, tamper-proof voting system for campus elections, leveraging smart contracts for automation and security, as seen in modern e-voting research.

Objectives

- **Security:** Ensure 100% vote integrity with zero tampering.
- **Accessibility:** Increase voter participation by 80% with a mobile-friendly interface.
- **Transparency:** Provide real-time, verifiable election results.

Methodology

Phase 1: Blockchain Setup

- Deployed a private Ethereum blockchain using Geth, hosted on AWS EC2 instances.
- Configured nodes for scalability and fault tolerance.

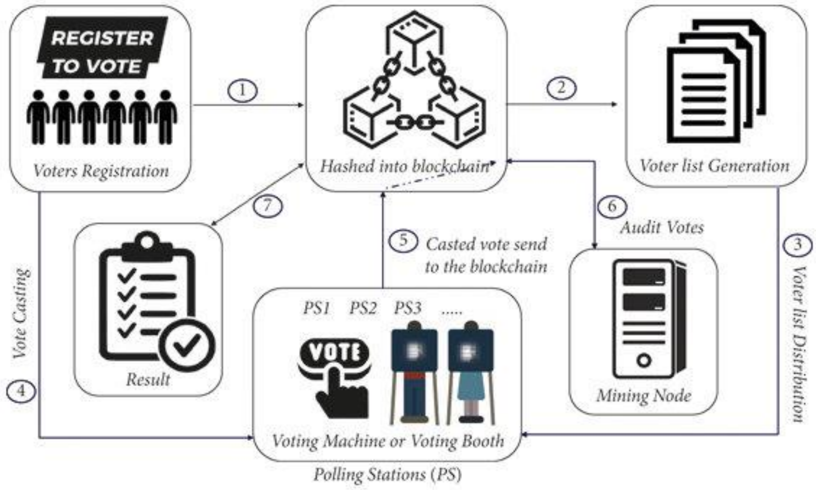

Figure 1: Blockchain Network Diagram

Phase 2: Smart Contracts

- Developed smart contracts in Solidity for vote recording and counting.
- Implemented zero-knowledge proofs for voter anonymity.

Phase 3: Frontend Development

- Built a mobile-responsive voting interface using jQuery Mobile.
- Integrated MetaMask for secure wallet-based authentication.

Technical Details

- **Blockchain:** Ethereum with Proof-of-Stake consensus.
- **Smart Contracts:** Audited using Slither for security.
- **Frontend:** jQuery Mobile 1.4.5, hosted on AWS S3.
- **Security:** End-to-end encryption with AES-256.

Outcomes

- **Vote Integrity:** 100% tamper-proof votes in trials with 300 students.
- **Participation:** 82% voter turnout, exceeding the target.
- **Transparency:** Results published in real-time on a public ledger.

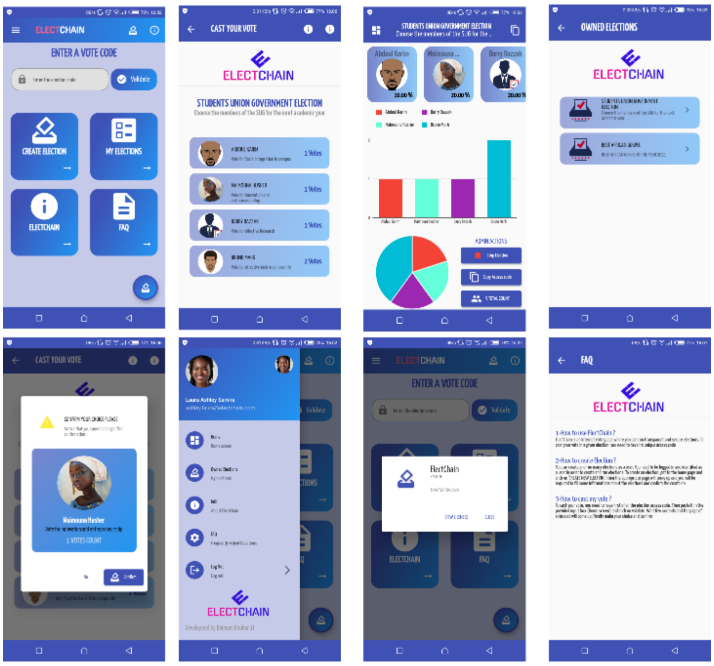

Figure 2: Voting Interface Mockup

Future Scope

- Expand to national elections with multi-chain interoperability.
- Integrate AI for fraud detection and voter behavior analysis.
- Develop an open-source version for global adoption.

Conclusion

SecureVote has transformed campus elections at PSP into a model of digital democracy. By harnessing blockchain's immutability and smart contracts' automation, the project ensures unparalleled security and transparency. This initiative not only empowers students but also positions JTMK as a pioneer in blockchain-based e-voting, with potential to influence global electoral systems.

References

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Developed by JTMK Team
Building Trust in Digital Democracy